LESSON PLAN · SCRATCH (GR 3–5) INTERMEDIATE: STORY, QUIZ & MUSIC STUDIO

Lesson 1: Loops Inside Loops: Pattern Art

Module 1 | 45 minutes | Nested loops | Grades 3–5 · ages 8–10

Objective: Use a loop inside another loop (a nested loop) to draw repeating patterns such as flowers and rows of shapes.

Success criteria

- I can build a loop inside a loop.
- I can work out how many times the inner blocks run.
- I can space shapes evenly around a circle.

Key vocabulary

- **nested loop**: A loop inside another loop.
- **inner loop**: The loop inside; it runs completely each time.
- **outer loop**: The loop on the outside that repeats the inner loop.
- **pen size**: How thick the pen line is.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Clap-stamp nests (5 min): “Clap 3 times” is the inner loop; “do the clapping 4 times, stamping once after each” is the outer loop. Count the total claps together (12).
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Draw a flower of squares: learners follow the steps.
14 min	**Perform**	Code Lab: flower of squares, Count the work, Loop maths
9 min	**Probe**	Code Lab: row of squares, activity
4 min	**Wrap-up**	Exit ticket: Outer loop 5, inner loop 4: how many times does the move block run? How did you work it out?

Unplugged warm-up

Clap-stamp nests (5 min): “Clap 3 times” is the inner loop; “do the clapping 4 times, stamping once after each” is the outer loop. Count the total claps together (12).

Teaching the idea

A nested loop is a loop placed inside another loop. For every single repeat of the outer loop, the inner loop runs completely. The total number of runs is the product of the two counts. In pen art, the inner loop draws a shape and the outer loop repeats and rotates it.

Analogy: Like a clock: the minute hand goes round 60 times (inner) for every one turn of the hour hand (outer).

Live demo: Draw one square live. Ask: “How could we make 8 of these around a circle?” Wrap it in repeat 8 without the turn first (all squares on top of each other!), then add the 45° turn.

Questions to ask

Ask	Listen for
How many times will the inner blocks run?	Outer count × inner count.
Why did all the squares draw on top of each other?	There’s no turn in the outer loop.
How do you choose the outer turn?	360 ÷ number of shapes.

Differentiation

Support
Give learners the square script ready-made; they add only the outer loop and turn.

Stretch
Make a 12-petal flower of triangles with a different colour each petal, or a 3 × 3 grid of squares (two nested loops plus a third for rows).

Common misconceptions

Misconception	How to fix it
Nested loop counts add up (3 + 4)	They multiply: 3 × 4 = 12.
The outer turn goes inside the inner loop	The inner loop draws one shape; the turn between shapes goes after it, inside the outer loop.
Pen up deletes lines	Pen up just stops drawing; erase all clears the stage.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Outer loop 5, inner loop 4: how many times does the move block run? How did you work it out?

Homework
Draw on paper what this makes: repeat 6 { repeat 3 { move 50, turn 120 }, turn 60 }.

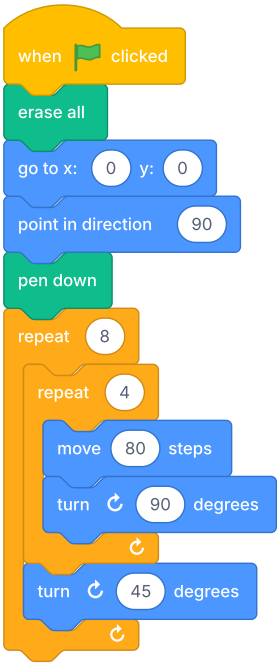
Answer key — Kitabash activities

1. Code Lab: flower of squares (perform · code lab)

Goals checked: Uses a loop inside a loop · Draws 8 squares

One solution:

Pip



2. Count the work (perform · multiple choice)

Answer: 12 times

3. Loop maths (perform · fill blank)

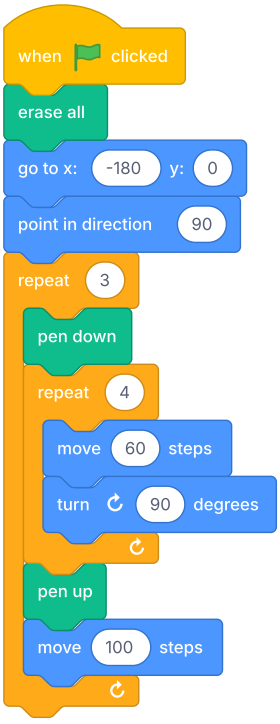
Answers: 30 · 60

4. Code Lab: row of squares (probe · code lab)

Goals checked: Uses a nested loop · Draws 3 squares (12 sides) · Lifts the pen between squares

One solution:

Pip



5. Explain your thinking (probe · free response)

Look for: inner, outer, times. **Example:** A school week is like a nested loop. The inner loop is the 6 lessons in a day, and the outer loop repeats the day 5 times. So I have 5 × 6 = 30 lessons.

Answer key — worksheet

1. **Key words:** nested loop = A loop inside another loop. · inner loop = The loop inside; it runs completely each time. · outer loop = The loop on the outside that repeats the inner loop. · pen size = How thick the pen line is.
2. **Plan your code:** Goals: Uses a loop inside a loop; Draws 8 squares.
3. **Count the work:** A — The inner loop runs 4 times, and the outer loop runs it 3 times: 3 × 4 = 12.
4. **Loop maths:** 30 · 60
5. **Plan your code:** Goals: Uses a nested loop; Draws 3 squares (12 sides); Lifts the pen between squares.

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